

Rethinking TOD's Reach on Housing Price

From Calgary to Montgomery County

Kwangyul Choi, Ph.D., Associate Professor
Department of Geoinformatic Engineering
Inha University

The International Joint Symposium & Workshop on Transit-Oriented Development 2026

Outline

1

TOD Overview

2

Case Study of Calgary, CANADA

3

Theoretical Background

4

The Research Gaps

5

New Study Design

6

Conclusions + Q&As

■ Definition

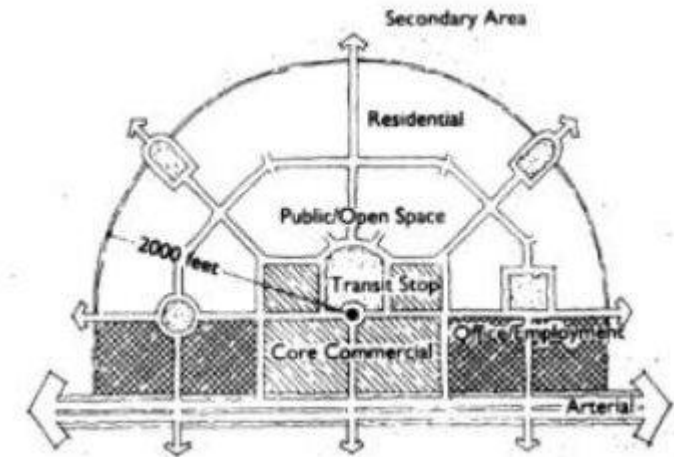
- Definition of Transit-Oriented Development

Definition

- Definition of Transit-Oriented Development
 - A Transit-Oriented Development (TOD) is **a mixed-use community within an average 2,000-foot walking distance of a transit stop and core commercial area**. TODs mix residential, retail, office, open space, and public uses in a walkable environment, making it convenient for residents and employees to travel by transit, bicycle, foot, or car.

Why do we draw a circle around every transit station?

- Definition of Transit-Oriented Development
 - A Transit-Oriented Development (TOD) is a **mixed-use community within an average 2,000-foot walking distance of a transit stop and core commercial area**. TODs mix residential, retail, office, open space, and public uses in a walkable environment, making it convenient for residents and employees to travel by transit, bicycle, foot, or car.



The size of a TOD must be determined on a case-by-case basis. **The average 2,000-foot radius** is intended to represent a “comfortable walking distance” (± 10 minutes) for a majority of people.

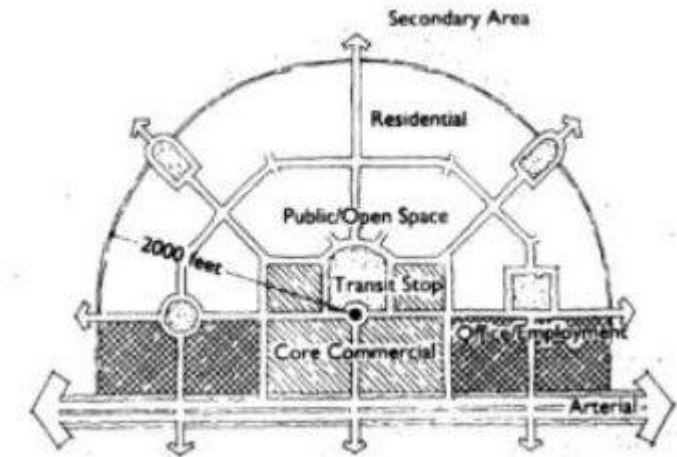
Why do we draw a circle around every transit station?

- Definition of Transit-oriented Development

1.0 INTRODUCTION

1.1 Transit Oriented Development Definition

Transit Oriented Development (TOD) is a walkable, mixed-use form of development typically focused within a 600m radius of a Transit Station – a Light Rail Transit (LRT) station or Bus Rapid Transit (BRT) stop prior to the arrival of LRT. Higher density development is concentrated near the station to make transit convenient for more people and encourage ridership. This form of development utilizes existing infrastructure, optimizes use of the transit network and creates mobility options for transit riders and the local community. Successful TOD provides a mix of land uses and densities that create a convenient, interesting and vibrant community for local residents and visitors alike.



The size of a TOD must be determined on a case-by-case basis. **The average 2,000-foot radius** is intended to represent a “comfortable walking distance” (± 10 minutes) for a majority of people.

Transit Oriented Development (TOD)

Transit Oriented Development (TOD) is an intentional mixing of land use and transit through the creation of compact, walkable, mixed-use communities within walking distance of a transit stop or station.

The goal of a TOD is to bring together people, jobs, and services designed in such a way to make it efficient, safe, and convenient to travel on foot or by bicycle, transit, or car. TOD is an opportunity for Austin to carefully coordinate transit and development for the benefit of the entire City. As Austin looks ahead over the next 20 years with the help of the *Imagine Austin Comprehensive Plan 2*, Transit Oriented Development is one tool for growing as a compact and connected city and developing as an affordable and healthy community. The City of Austin is committed to creating transit-supportive communities by optimizing land use around high quality transit.

TOD Principles (Institute for Transportation & Development Policy)

- 1) Walk
- 2) Cycle
- 3) Connect
- 4) Transit
- 5) Mix
- 6) Density
- 7) Compact
- 8) Shift



Illustration of "TOD" from Institute of Transportation and Development Policy

TOD Principles (Austin Planning)

TOD Principles

The following principles provide an understanding of the essential elements and characteristics of Transit Oriented Development (TOD). They serve as the foundation for the Station Area Planning process and implementation:

- Create compact development with sufficient density to support transit ridership located within easy walking distance of transit stops.
- Promote development strategies that focus on accommodating people on foot without excluding people traveling by vehicle.
- Encourage active and livable places that serve our daily needs and provide people with a sense of belonging and ownership within their community.
- Include engaging, high quality public spaces, such as small parks or plazas, as organizing features and gathering places for the neighborhood.
- Encourage a variety of housing choices near transit facilities to accommodate a wide range of ages and incomes.
- Incorporate retail into development, if viable at a particular location, ideally drawing customers from within the TOD and from major streets.
- Ensure compatibility and connectivity with surrounding neighborhoods.
- Introduce creative parking strategies to allow the accommodation of vehicles without dividing a site such that parking is the dominating design factor.
- Strive to make TODs realistic yet economically viable and valuable from a diversity of perspectives (city, transit agency, developer, resident, employer).
- Recognize that all TODs are not the same; each development is located within its own unique context and serves a specific purpose in the larger context.

TOD Goals

- The goal of a TOD is to bring together people, jobs, and services designed in such a way to make it efficient, safe, and convenient to travel on foot or by bicycle.
- Successful TOD leverages public investments in transit to promote:



Neighborhood revitalization



More affordable housing



Public- and private investment



Economic returns to surrounding landowners and businesses



Increased ridership for transit systems



Congestion relief and improved air quality



Improved safety for pedestrians and cyclists.



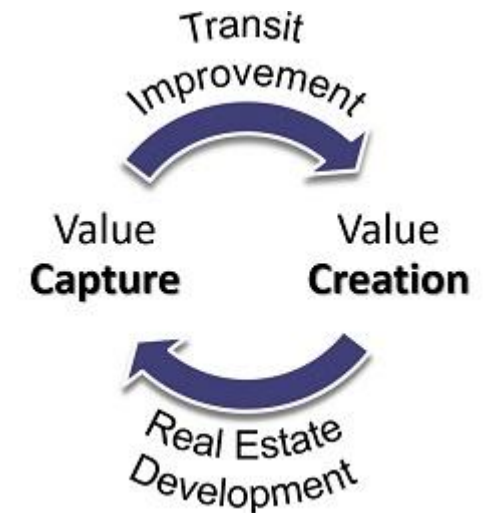
- 800 apartments (141 affordable)
- 120,000 ft² of office
- 110,000 ft² of retail

Value Capture

- Definition
 - A public financing strategy that recovers a share of the value created by transit infrastructure, used to help repay construction debt and fund ongoing operations and maintenance
- Key Mechanisms
 - Tax incremental financing, special assessments, and joint development
- Findings from Studies
 - Transit project increase nearby property values by 30-40%, and by as much as 150% under ideal conditions

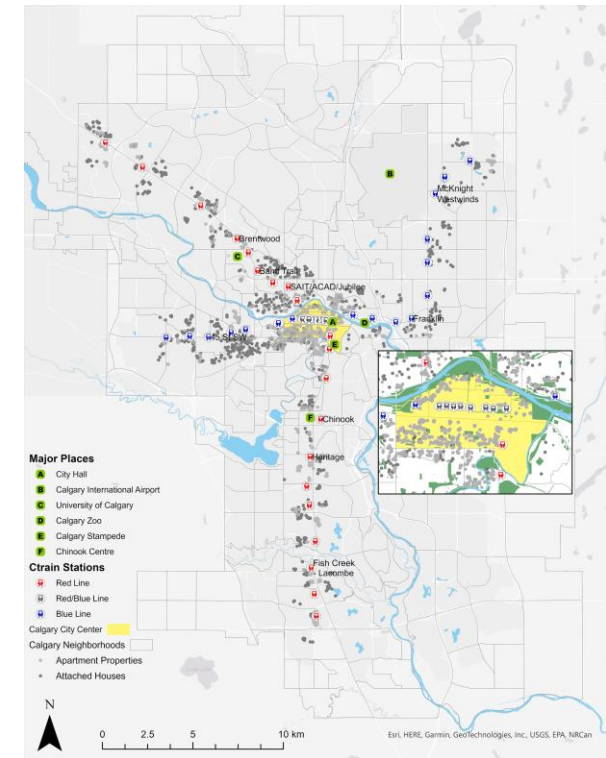


Plaza Saltillo Joint Development Project (Left) & Kansas City Streetcar Transportation Development District (Right)



Case Study of Calgary, CANADA

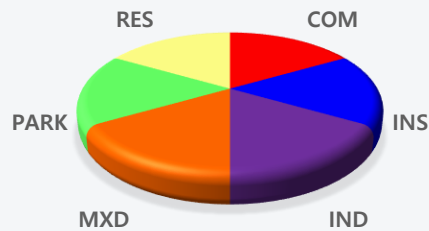
- Choi, K., Park, H. J. & Alaniz Uribe, F. (2023). The impact of light rail transit station area development on residential property values in Calgary, Canada: Focus on land use diversity and activity opportunities. *Case Studies on Transport Policy*, 12, 100924.



Case Study of Calgary, CANADA

- Research Question
 - How much will a home buyer pay for mixed-use development (land use diversity) of LRT station?
- Study Area
 - 46 LRT stations
 - Non-detached residential properties within 1,500m of stations
- Two TOD Measures

Land Use Entropy



$$0 \leq \text{Entropy} \leq 1$$

Activity Opportunity Count

Calgary Business License
(CBL)



Structural Features,
Transit Accessibility &
Neighborhood Characteristics

- Multi-level Hedonic Price Model
 - Attached houses & apartment properties

■ Case Study of Calgary, CANADA

- Findings

Variable	Attached Houses	Apartments
Land use entropy		
Activity opportunities		

■ Case Study of Calgary, CANADA

- Findings

Variable	Attached Houses	Apartments
Land use entropy	Insignificant	Insignificant
Activity opportunities		

■ Case Study of Calgary, CANADA

- Findings

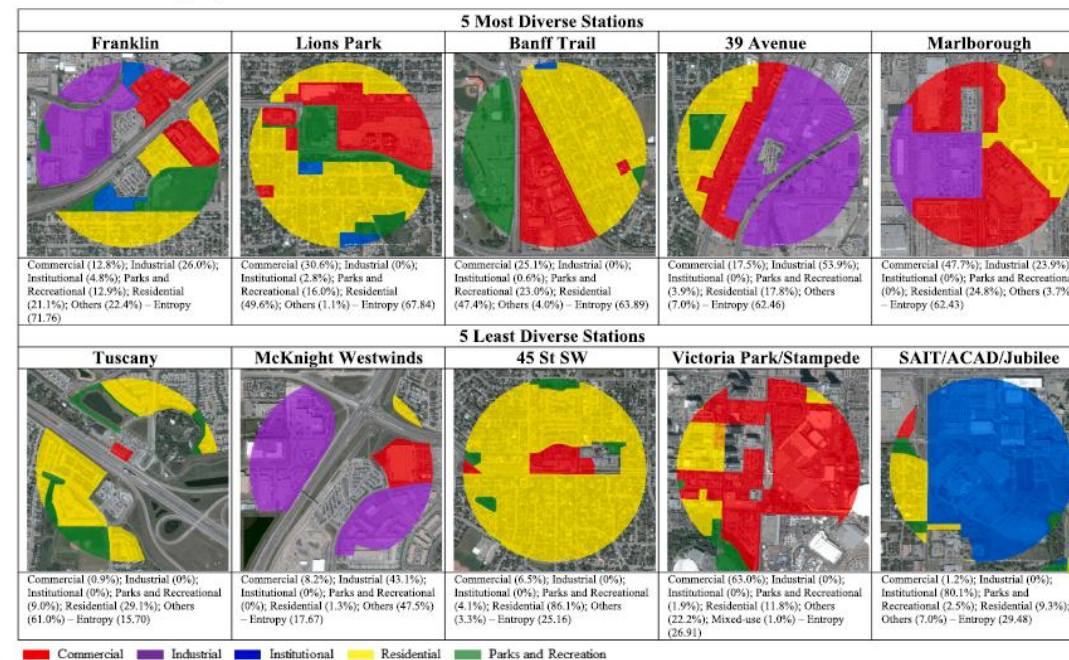
Variable	Attached Houses	Apartments
Land use entropy	Insignificant	Insignificant
Activity opportunities	Insignificant	<i>Significant, positive</i>

Case Study of Calgary, CANADA

Findings

Variable	Attached Houses	Apartments
Land use entropy	Insignificant	Insignificant
Activity opportunities	Insignificant	<i>Significant, positive</i>

Table 3
Station Land Use Entropy (Top 5 & Bottom 5).



Case Study of Calgary, CANADA

Findings

Variable	Attached Houses	Apartments
Land use entropy	Insignificant	Insignificant
Activity opportunities	Insignificant	<i>Significant, positive</i>

Table 3
Station Land Use Entropy (Top 5 & Bottom 5).

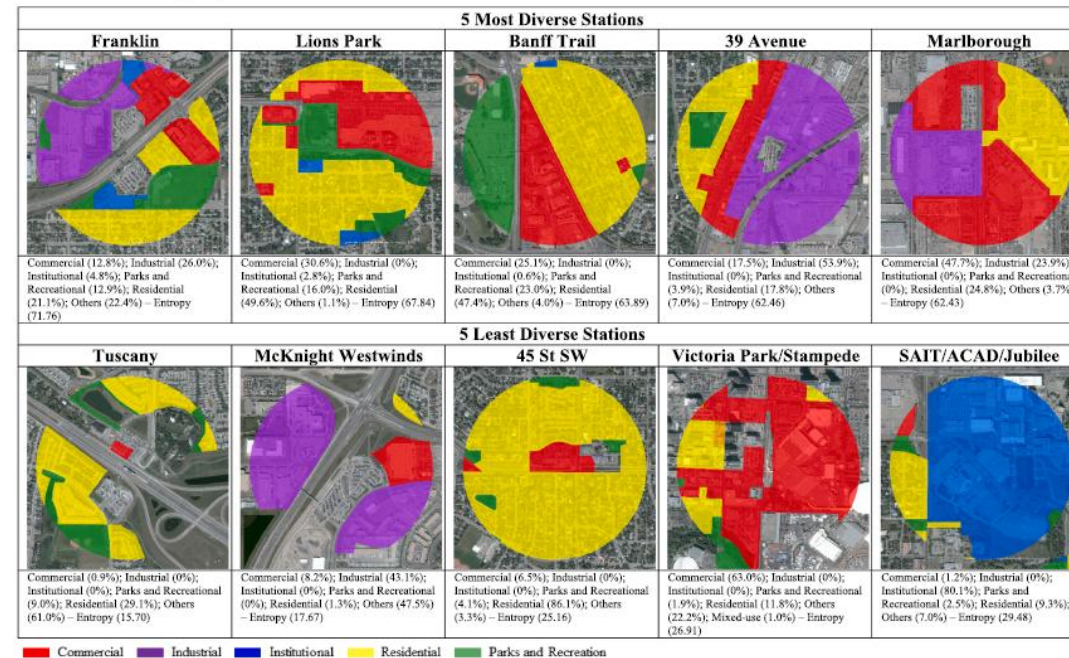


Table 4
Number of Activity Opportunities around Station.

Rank	Station Name	Activity Count	Rank	Station Name	Act Count
8	Marlborough	229	29	Sunalta	29
9	Rundle	173	30	McKnight Westwinds	26
10	Chinook	136	31	Fish Creek - Lacombe	25
11	Sunnyside	130	31	Canyon Meadows	25
14	Lions Park	111	31	Shawnessy	25
15	Westbrook	103	34	Saddletown	24
16	Victoria Park / Stampede	95	35	Bridgeland / Memorial	23
17	Downtown West - Kerby Station	81	36	Whitehorn	14
18	Southland	78	37	SAIT/ACAD/ Jubilee	12
19	Dalhousie	76	38	Erlton / Stampede	6
20	39 Avenue	73	39	Barlow / Max Bell	4
21	Sirocco	57	40	69 St SW	3
22	Somerset-Bridlewood	51	41	45 St SW	1
23	Banff Trail	46	42	Shaganappi Point	0
24	Heritage	45	42	Martindale	0
25	Brentwood	44	42	Zoo	0
26	Franklin	37	42	University	0
27	Anderson	33	42	Tuscany	0
28	Crowfoot	30			

Case Study of Calgary, CANADA

- Findings
 - Unsubstantial mixed-use (Fig. 2)
 - Franklin Station (Entropy 0.83)
 - No destinations for pedestrians or transit users (Fig. 3)
 - Banff Trail Station (Entropy 0.6)
 - Uncomfortable access (Fig. 4)
 - Franklin Station
 - Brentwood Station



Fig. 2. Franklin Station Note: Church (Left), Commercial (Middle), Parking Lot (Right).



Fig. 3. Banff Trail Station. Note: Motels (Left), Car dealership (Middle), Pedestrian overpass to the stadium (Right).



Fig. 4. Grade-separated Stations. Note: Franklin Station (Left) and Brentwood Station (Right).

Case Study of Calgary, CANADA

- Findings
 - Unsubstantial mixed-use (Fig. 2)
 - No destinations for pedestrians or transit users (Fig. 3)
 - Uncomfortable access (Fig. 4)



Fig. 2. Franklin Station Note: Church (Left), Commercial (Middle), Parking Lot (Right).



Fig. 3. Banff Trail Station. Note: Motels (Left), Car dealership (Middle), Pedestrian overpass to the stadium (Right).

Diverse land use $\neq$ Higher value

- ✓ What mattered was what you could actually do there!



Fig. 4. Grade-separated Stations. Note: Franklin Station (Left) and Brentwood Station (Right).

Case Study of Calgary, CANADA

- Limitations
 - **First**, we **did not consider pedestrian facilities** such as connections between places, sidewalks, and crosswalks of station areas as well as activities possibly happening in public spaces near stations such as wandering, sitting, and gathering.
 - **Second**, we **only accounted for land use from broad categories**. Future research would benefit from a further breakdown of subcategories and necessarily to look at a finer grain scale of land use. For example, commercial land use could be broken into retail, fast food, restaurant/ cafe, groceries, fitness services, professional services, to name a few.
 - **Third**, the impact of the proximity to the rail station may vary depending on the property type. We **did not include single-family homes** because of insufficient single-family home samples within the defined TOD areas in Calgary. However, other cities with a statistically meaningful number of single-family houses located close to light rail stations could complement our research by showing the influence of station area development on single-family homes, which is one of the most common housing types in North America.
 - **Finally**, the sale transaction **data were from the pre-pandemic period**, and so different results might be obtained from the same analysis using data collected during the post-pandemic period.

Theoretical Backgrounds

Two Theoretical Pillars

Hedonic Price Model

- Rent or price is a bundle of implicit prices for the property's individual attributes – structural (S), locational (L), and environmental (E).

$$P_i = f(S, L, E)$$

TOD 1.0

The five elements of a prototype TOD

- A transit station – the anchor of the district
- A walking range district – ¼ - ½ miles
- Medium to high density development
- Mixed land uses
- Pedestrian-centered environmental design



Efficiency;
Equity;
Eco-adaptivity

■ Research Gaps

- Three Gaps, One New Study

Gaps from Calgary	New Study's Responses
Single-family homes excluded	County with abundant single-family housing stock
One fixed-distance (e.g., 400m) buffer for everything	Let the data find each variable's own scale
One coefficient for the whole study	Let effects vary locally, station by station

Research Gaps

■ Three Gaps, One New Study

Gaps from Calgary	New Study's Responses
Single-family homes excluded	County with abundant single-family housing stock
One fixed-distance (e.g., 400m) buffer for everything	Let the data find each variable's own scale
One coefficient for the whole study	Let effects vary locally, station by station

Core Questions:

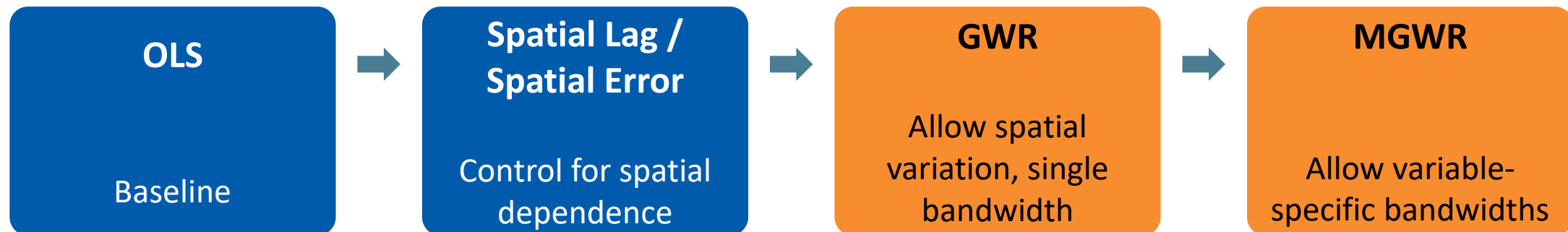
- 1. How far from the station does each TOD dimension's influence actually extend?***
- 2. Does that price effect vary by location across the region?***
- 3. Does the SCALE of that variability differ across the TOD dimensions?***

Study Design Overview

Building the variable framework

Origin	Variable	Method
Transit station	Distance to transit	Point-based
Medium-to-high density development	Density	Station-anchored, decay-weighted
Mixed land uses	Diversity	Station-anchored, decay-weighted
Pedestrian-centered design	Design	Station-anchored, decay-weighted
Destination accessibility	Destination accessibility	Point-based

Building the modeling framework



The Core Idea: Bandwidth as Catchment Area

- Coefficient Surfaces of OLS, GWR, and MGWR
 - OLS model
 - Constant across the space
 - GWR model
 - The relationships change spatially, but each variable changes according to the same bandwidth.
 - MGWR model
 - The relationships change at a different rate – the bandwidths increase from left to right (Figure 1).

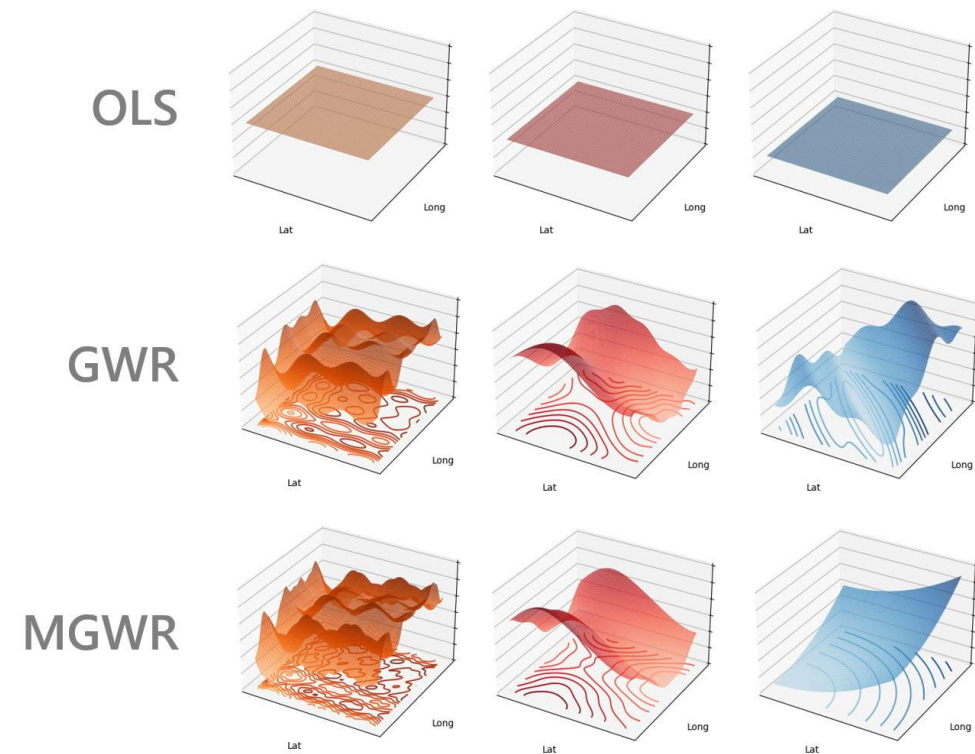


Figure 1: coefficient surfaces of OLS, GWR, and MGWR models

■ Conclusions

- Summary
 - Calgary asked *what matters* near a station
 - The new study asks *how far* does it matter

Conclusions

- Summary
 - Calgary asked *what matters* near a station
 - The new study asks *how far* does it matter
- Question to carry with you
 - At a station you visit in Korea, does it feel as mixed and walkable on the ground? If not, which breaks down first—density, diversity, or design—and about how far from the station does it happen?

Thank You